

Discrete Mathematics

Week 12: Graphs Part 1

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Graphs

A graph represents connections between objects.

- Acquaintance between people
- Collaboration between working people
- Links between websites
- Computer network and the Internet
- Road and rail network

Chapter Summary

1. Graphs and Graph Models
2. Graph Terminology and Special Types of Graphs
3. Representing Graphs
4. Connectivity
5. Euler and Hamiltonian Graphs

Graphs

Definition:

A **graph** G consists of V , a nonempty set of **vertices** (or **nodes**) and E , a set of **edges**.

- This can be written as $G = (V, E)$
- Each edge connects either one or two vertices associated with it, called **endpoints**.

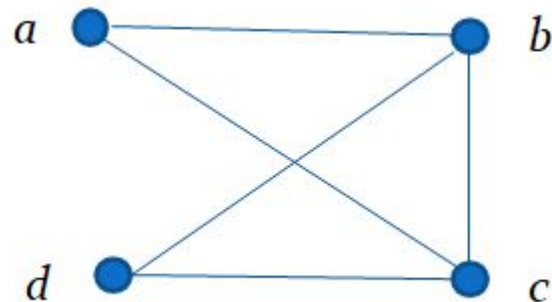
Graphs

A graph with an infinite vertex set is called an *infinite graph*.

A graph with a finite vertex set is called a *finite graph*.

Example: A graph with 4 vertices and 5 edges.

- $V = \{a, b, c, d\}$
- $E = \{\{a, b\}, \{a, c\}, \{b, c\}, \{b, d\}, \{c, d\}\}$



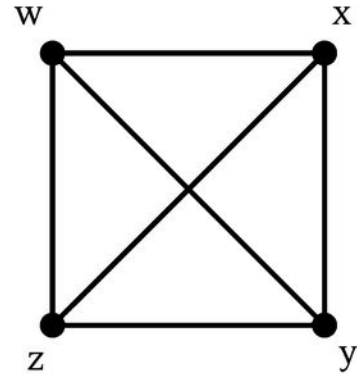
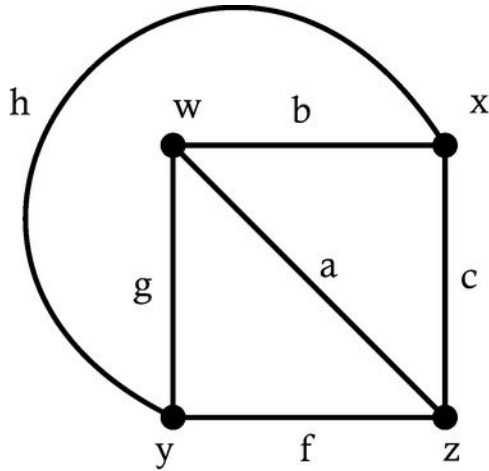
1. Graphs and Graph Models

There are different kinds of graphs: directed or undirected, and loops.

Undirected Graph Terminology

In a **simple graph**:

- Each edge connects two different vertices
- No two edges connect the same pair of vertices.



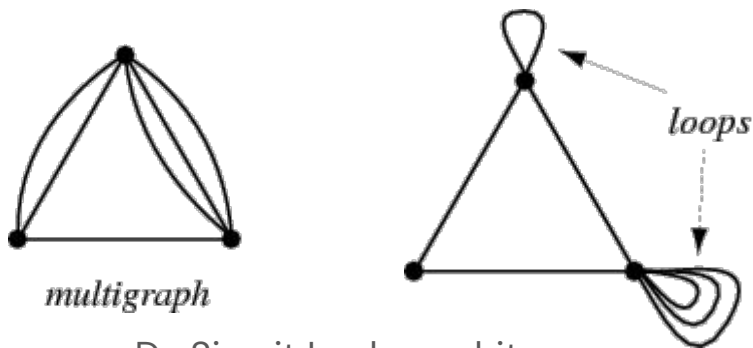
Undirected Graph Terminology

Multigraphs may have multiple edges connecting the same two vertices.

- When m different edges connect the vertices u and v , we say that $\{u, v\}$ is an edge of multiplicity m .

Loop is an edge that connects a vertex to itself.

Pseudograph may include both loops and multiple edges.



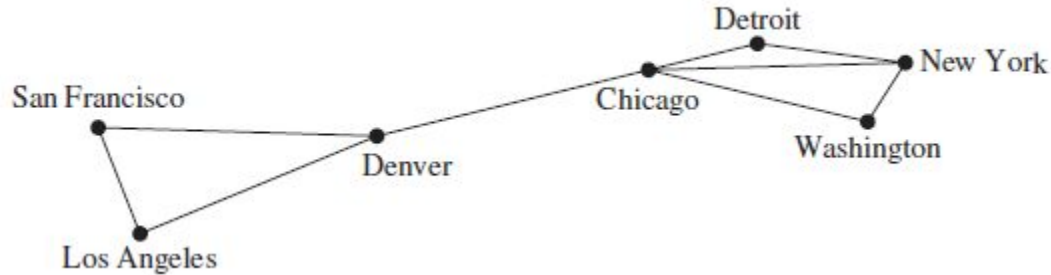
Graph Models: Computer Networks

A network is made up of data centers and communication links between computers.

In all these graph models,

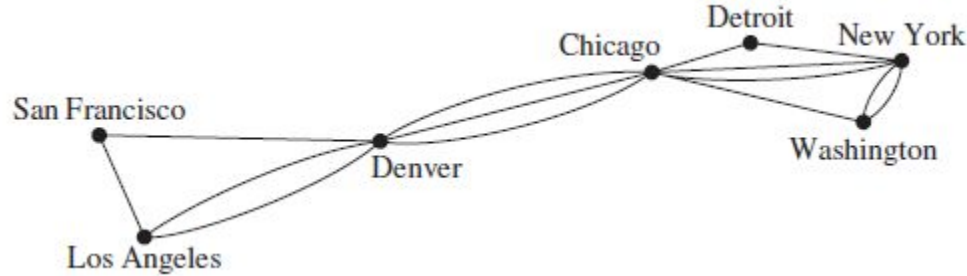
- the vertices represent data centers
- the edges represent communication links.

Graph Models: Computer Networks



Simple Graph

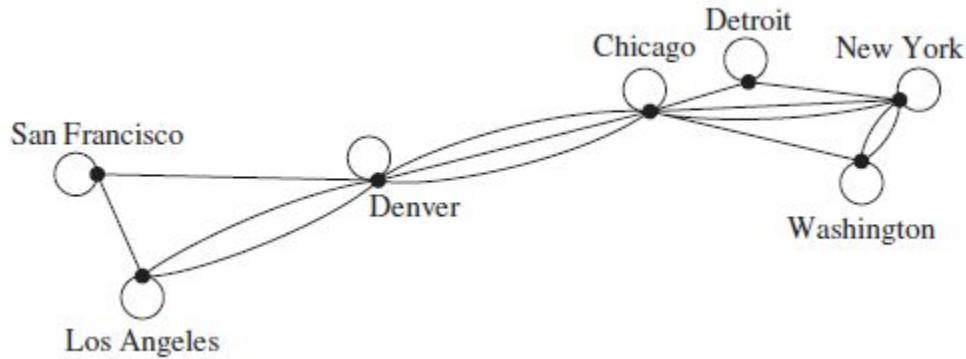
Graph Models: Computer Networks



Multigraph

- A multigraph can be used to model a computer network where we care about the number of links between data centers.

Graph Models: Computer Networks



Pseudograph:

Multigraphs with loops

- To model a computer network with diagnostic links at data centers, we use a **pseudograph**, as feedback loops are needed.

Directed Graph

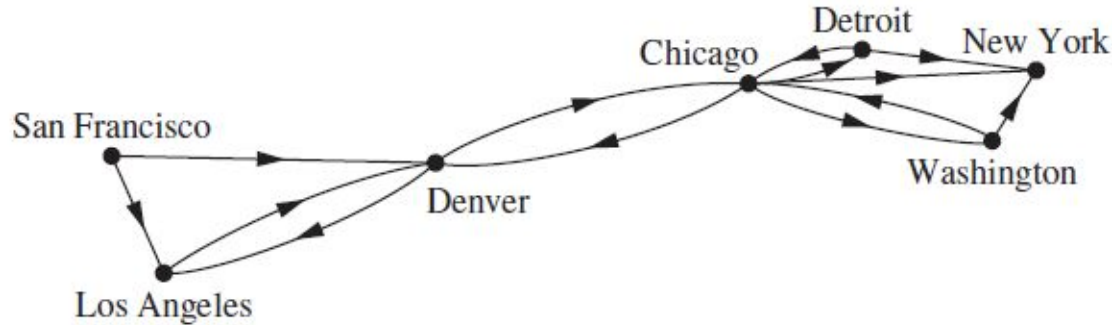
Definition:

A **directed graph** (or **digraph**) $G = (V, E)$ where

- Each edge is associated with an ordered pair of vertices.
- The directed edge associated with the ordered pair (u, v) is said to start at u and end at v .

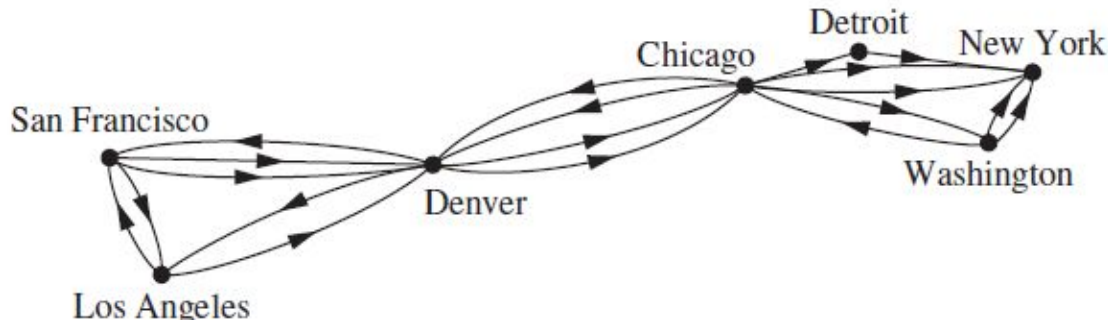
Graphs where the endpoints of an edge are not ordered are said to be **undirected graphs**.

Graph Models: Computer Networks



Simple Directed Graph

- To model a network with direction.

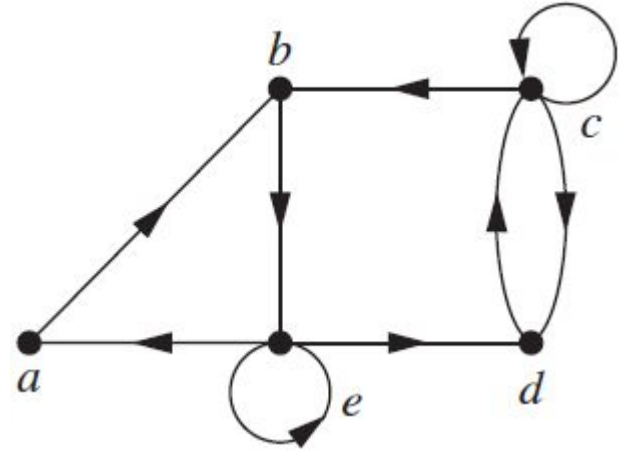
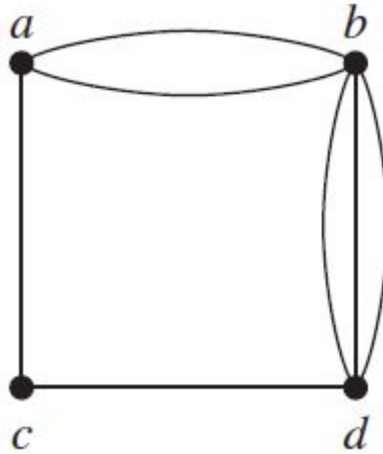
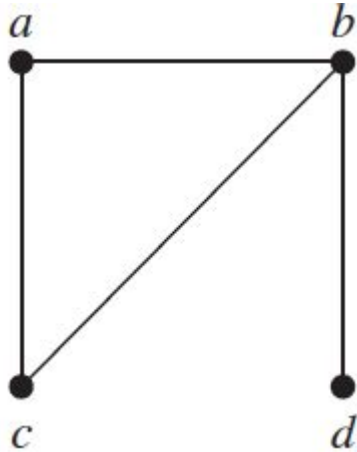


Directed Multigraphs

- To model a network with multiple one-way links.

Type of Graph Exercise

Simple graph, multigraph, directed?



Graph Exercise

Exercise 1: Draw graph models, stating the type of graph used, to represent airline routes where every day there are

- 4 flights from Bangkok to Chiang Mai,
- 2 flights from Chiang Mai to Bangkok,
- 3 flights from Chiang Mai to Krabi,
- 2 flights from Krabi to Chiang Mai,
- 1 flight from Chiang Mai to Loei,
- 2 flights from Loei, to Chiang Mai,
- 3 flights from Chiang Mai to Samui,
- 2 flights from Samui to Chiang Mai, and
- 1 flight from Samui to Krabi, with (*continued in next page*)

Graph Exercise

- a) An edge between vertices representing cities that have a flight between them (in either direction).

Graph Exercise

- b) An edge between vertices representing cities for each flight that operates between them (in either direction).

Graph Exercise

- c) An edge from a vertex representing a city where a flight starts to the vertex representing the city where it ends.

Graph Exercise

- d) An edge for each flight from a vertex representing a city where the flight begins to the vertex representing the city where the flight ends.

Other Applications of Graphs

- Social networks
- Communications networks (Call graphs)
- Information networks (Web pages and links)
- Software design (Modules/Library dependencies)
- Transportation networks (Airline routes and road networks)
- Biological networks (Species and protein interaction network)
- Semantic Networks (NLP)
- Tournaments

2. Graph Terminology and Special Types of Graphs

Undirected Graph Terminology

Definition 1: Two vertices u, v in an undirected graph G are called **adjacent** (or **neighbors**) in G if there is an edge e between u and v .

- Such an edge e is called **incident** with the vertices u and v and e is said to connect u and v .

Undirected Graph Terminology

Definition 2. The set of all neighbors of a vertex v of $G = (V, E)$, denoted by $N(v)$, is called the **neighborhood** of v .

- Neighbors: adjacent vertices

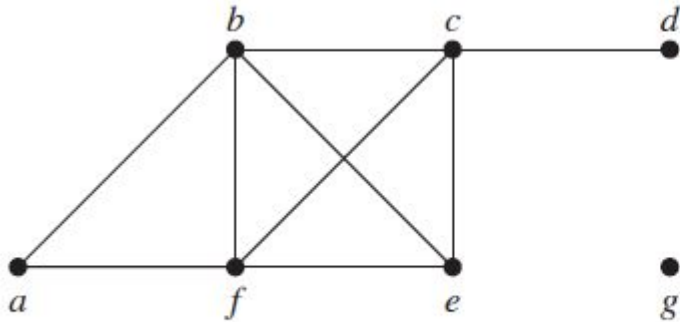
Undirected Graph Terminology

Definition 3: The **degree of a vertex** in a undirected graph is the number of edges incident with it, except that a loop at a vertex contributes two to the degree of that vertex.

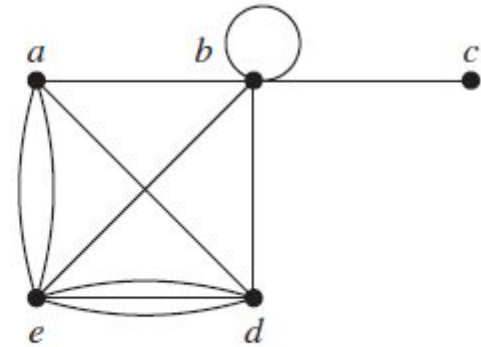
- The degree of the vertex v is denoted by $\deg(v)$.

Degrees and Neighborhoods of Vertices

Example: What are the degrees and neighborhoods of the vertices in the graphs G and H ?



G



H

Degrees of Vertices and Handshaking Theorem

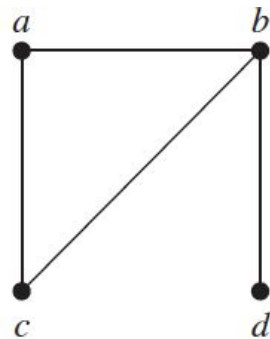
Theorem 1 (Handshaking Theorem): If $G = (V, E)$ is an undirected graph with m edges, then

$$2m = \sum_{v \in V} \deg(v)$$

Proof:

Each edge contributes twice to the degree count of all vertices.

Hence, both the left-hand and right-hand sides of this equation equal twice the number of edges.



Degrees of Vertices

Theorem 2: An undirected graph has an even number of vertices of odd degree.

Proof: Let V_1 be the vertices of even degree and V_2 be the vertices of odd degree in an undirected graph $G = (V, E)$ with m edges. Then

$$2m = \sum_{v \in V} \deg(v) = \sum_{v \in V_1} \deg(v) + \sum_{v \in V_2} \deg(v).$$

Handshaking Theorem Example

Example 1: How many edges are there in an undirected graph with 10 vertices of degree 6 for each vertex?

Example 2: If a graph has 5 vertices, can each vertex have degree 3?

Handshaking Theorem Example

Example 3: If a graph has 4 vertices, each have 1, 1, 3, 3 degrees respectively. Can an undirected graph be drawn?

Example 4: If a graph has 4 vertices, each have 2, 2, 2, 3 degrees respectively. Can an undirected graph be drawn?

Exercise

Exercise 1: If a graph has 4 vertices, each have 1, 2, 3, 4 degrees respectively. Can a simple undirected graph be drawn?

Exercise

Exercise 2: If a graph has 8 vertices, each have 1, 1, 2, 2, 9, 1, 3, 1 degrees respectively. Can a simple undirected graph be drawn?

Exercise

Exercise 3: Suppose there are 15 students, is it possible that each student will have at least 3 students as friend?

Exercise

Exercise 4: Suppose there are 6 people in a room, and each must shake hands with every other person. How many handshakes are there?

Directed Graph Terminology

Definition 4: When (u, v) is an edge of the graph G with directed edges, u is said to be **adjacent to** v and v is said to be **adjacent from** u .

- The vertex u is called the **initial vertex** of (u, v) , and v is called the **terminal** or **end vertex** of (u, v) .
- The initial vertex and terminal vertex of a loop are the same.

Directed Graph Terminology

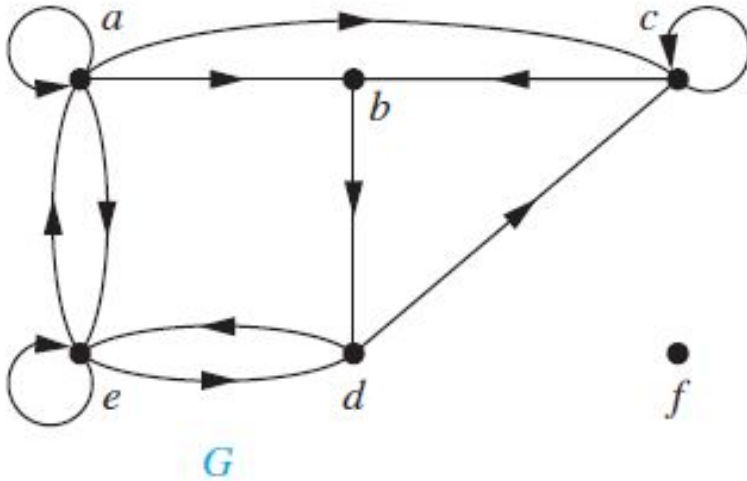
Definition 5: In a graph with directed edges the **in-degree of a vertex v** , denoted by $\deg^-(v)$, is the number of edges with v as their terminal vertex.

The **out-degree of v** , denoted by $\deg^+(v)$, is the number of edges with v as their initial vertex.

- Note that a loop at a vertex contributes 1 to both the in-degree and the out-degree of this vertex.

Directed Graph Example

Example: Find the in-degree and out-degree of each vertex in the graph G with directed edges.



Directed Graph

Theorem 3: Let $G = (V, E)$ be a graph with directed edges. Then

$$|E| = \sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v).$$

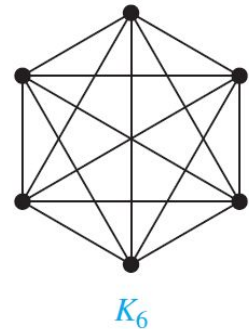
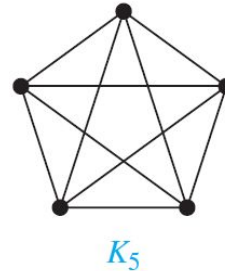
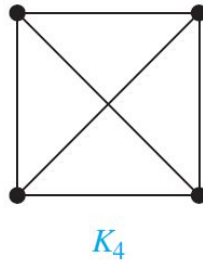
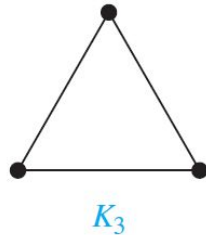
Proof:

- Because each edge has an initial vertex and a terminal vertex, the sum of the in-degrees and the sum of the out-degrees of all vertices in a graph with directed edges are the same.

Special Types of Simple Graphs

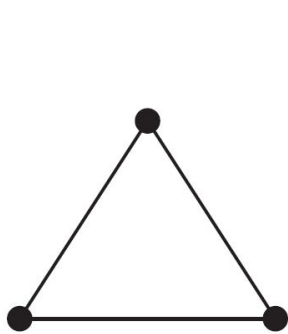
A **complete graph** on n vertices, denoted by K_n , is a simple graph that contains exactly one edge between each pair of distinct vertices.

A simple graph for which there is at least one pair of distinct vertex not connected by an edge is called **noncomplete**.

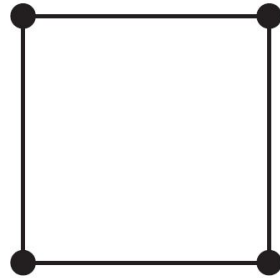


Special Types of Simple Graphs

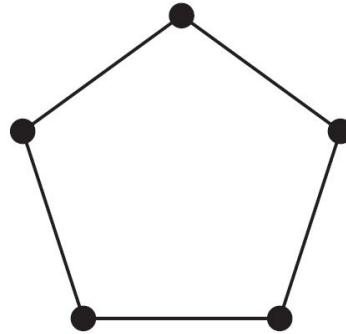
A **cycle** C_n , $n \geq 3$, consists of n vertices $v_1, v_2, \dots, v_n$ and edges $\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}$, and $\{v_n, v_1\}$.



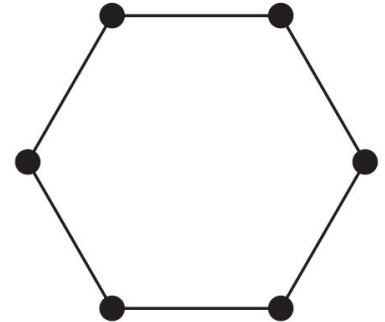
C_3



C_4



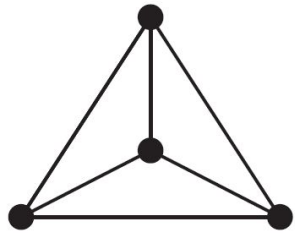
C_5



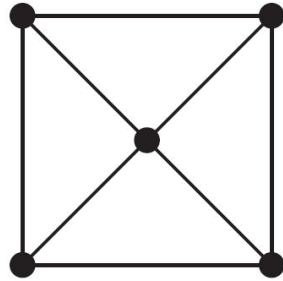
C_6

Special Types of Simple Graphs

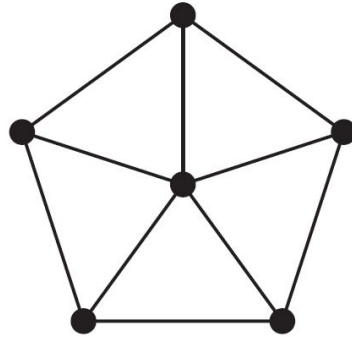
A **wheel** W_n is obtained by adding an additional vertex to a cycle C_n , for $n \geq 3$, and connect this new vertex to each of the n vertices in C_n by new edges.



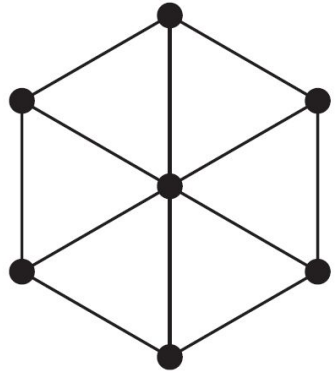
W_3



W_4



W_5



W_6

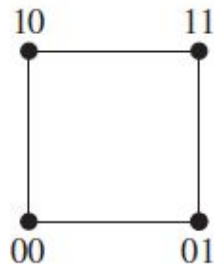
Special Types of Simple Graphs

An **n -dimensional hypercube**, or **n -cube**, denoted by $\mathbf{Q}_n$, is a graph that has vertices representing the 2^n bit strings of length n .

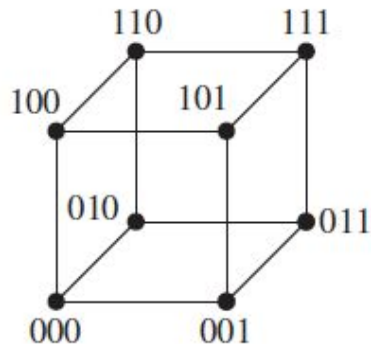
Two vertices are adjacent if and only if the bit strings that they represent differ in exactly one bit position.



Q_1



Q_2



Q_3

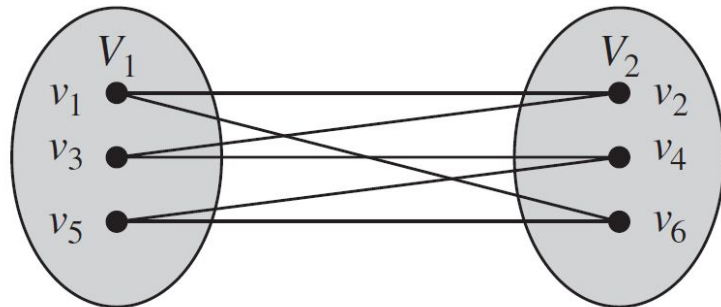
Bipartite Graphs

Definition 6: A simple graph is called **bipartite** if its vertex set V can be partitioned into 2 disjoint subsets V_1 and V_2 such that each edge connects a vertex from one subset to a vertex of the other.

- No edge connects either two vertices in V_1 or two vertices in V_2

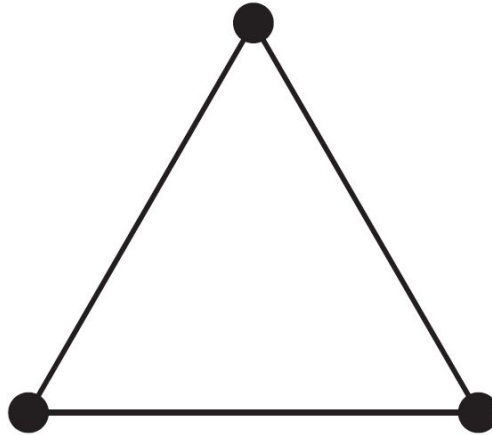


C_6



Example

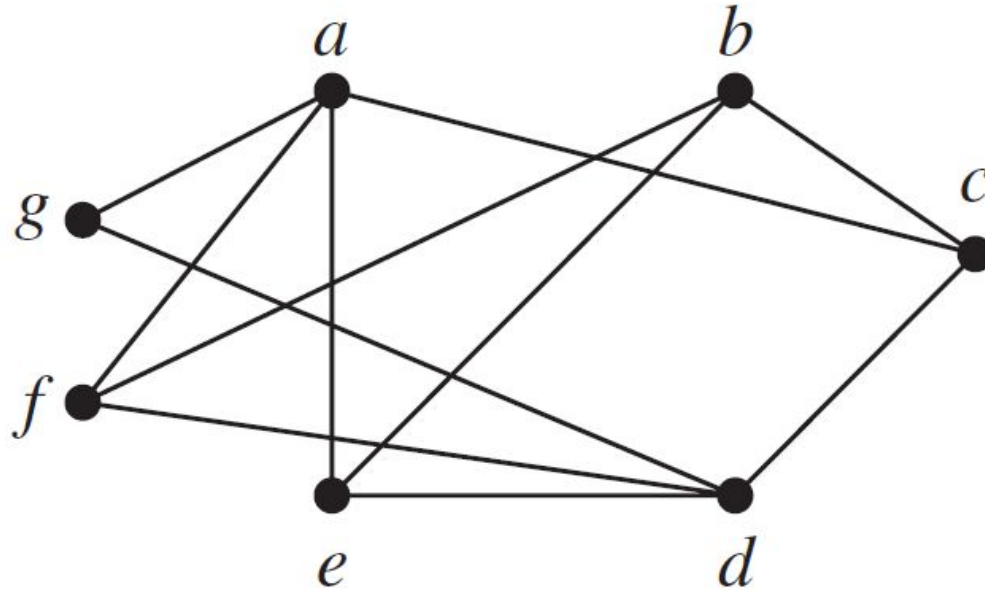
Example 5: Is this graph bipartite?



K_3

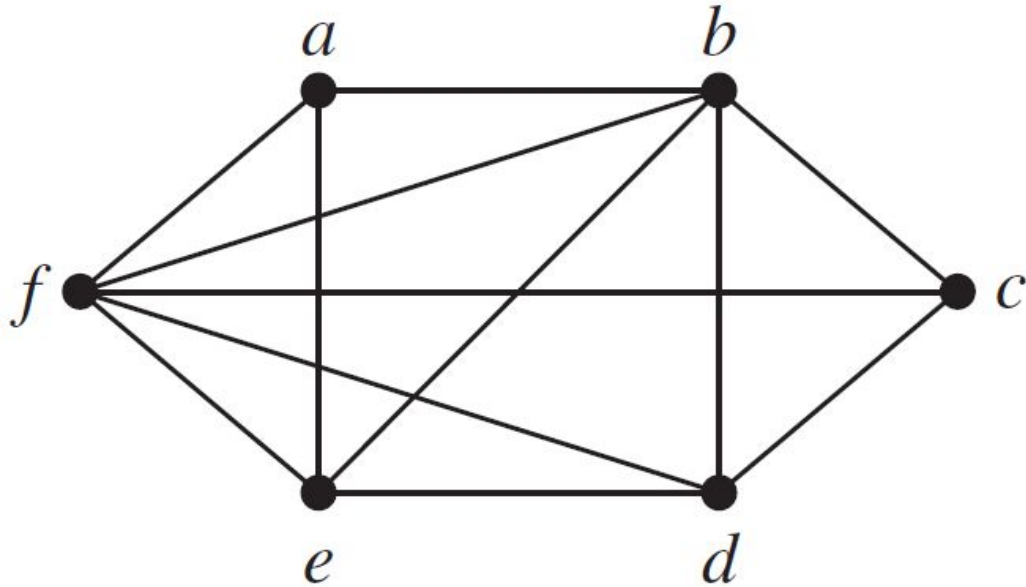
Exercise

Exercise 5: Is this graph bipartite?



Exercise

Exercise 6: Is this graph bipartite?

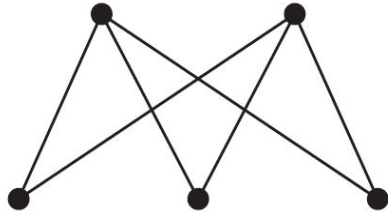


Complete Bipartite Graphs

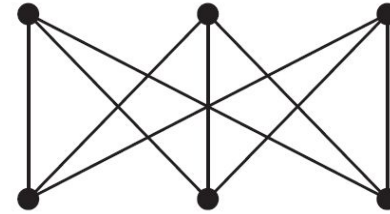
Definition 7: A **complete bipartite graph** $K_{m,n}$ is a bipartite graph with subsets of m and n vertices, respectively with an edge between each pair of vertices from opposite subsets.

- An edge between two vertices where one vertex is in the first subset and the other vertex is in the second subset.

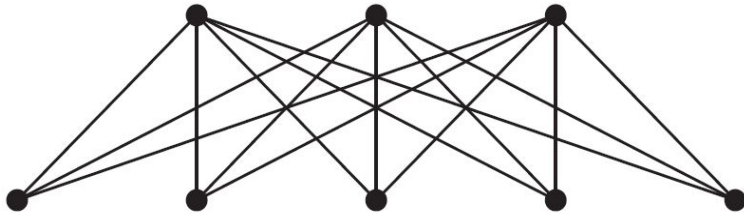
Complete Bipartite Graphs



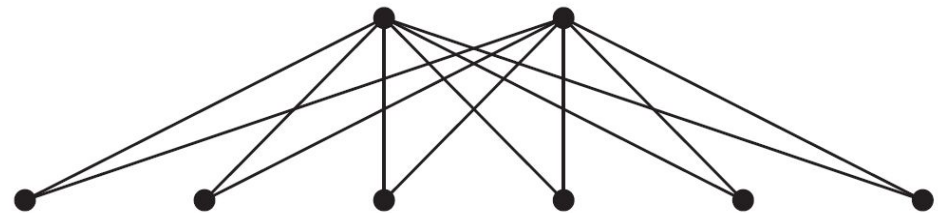
$K_{2,3}$



$K_{3,3}$



$K_{3,5}$



$K_{2,6}$

3. Representing Graphs

Adjacency Lists

Definition 1: An adjacency list can be used to represent a graph with no multiple edges by specifying the vertices that are adjacent to each vertex of the graph.

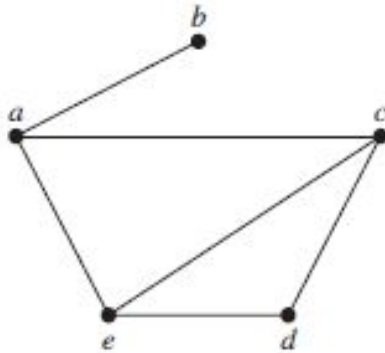


TABLE 1 An Adjacency List for a Simple Graph.

Vertex	Adjacent Vertices
<i>a</i>	<i>b, c, e</i>
<i>b</i>	<i>a</i>
<i>c</i>	<i>a, d, e</i>
<i>d</i>	<i>c, e</i>
<i>e</i>	<i>a, c, d</i>

Adjacency Lists Exercise

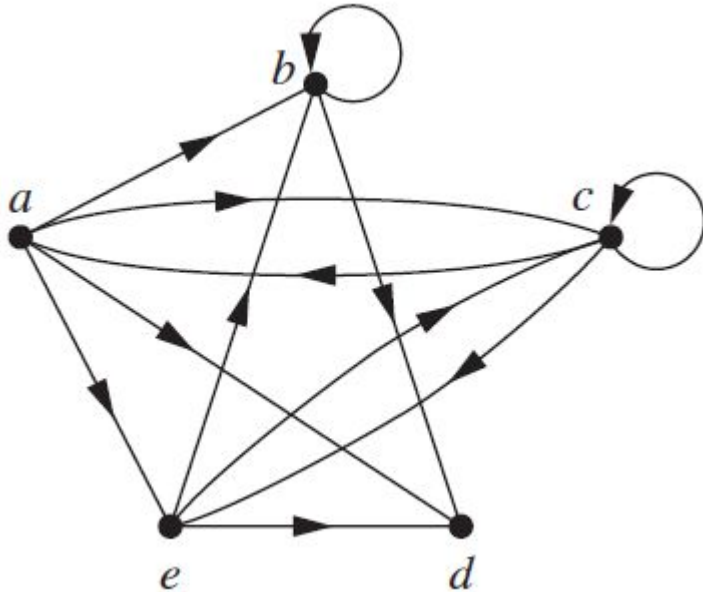


TABLE 2 An Adjacency List for a Directed Graph.

<i>Initial Vertex</i>	<i>Terminal Vertices</i>
<i>a</i>	
<i>b</i>	
<i>c</i>	
<i>d</i>	
<i>e</i>	

Adjacency Matrices

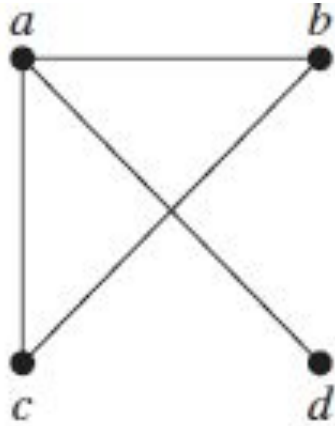
Definition 2: Suppose that $G = (V, E)$ is a simple graph where $|V| = n$. Suppose that the vertices of G are listed arbitrarily as $v_1, v_2, \dots, v_n$.

The adjacency matrix $\mathbf{A}_G$ of G , with respect to the listing of vertices, is the $n \times n$ zero-one matrix with 1 as its (i, j) th entry when v_i and v_j are adjacent, and 0 as its (i, j) th entry when they are not adjacent.

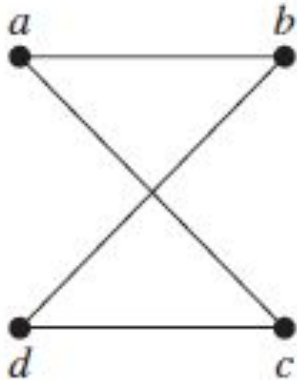
- In other words, if the graph's adjacency matrix is $\mathbf{A}_G = [a_{ij}]$, then

$$a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \text{ is an edge of } G, \\ 0 & \text{otherwise.} \end{cases}$$

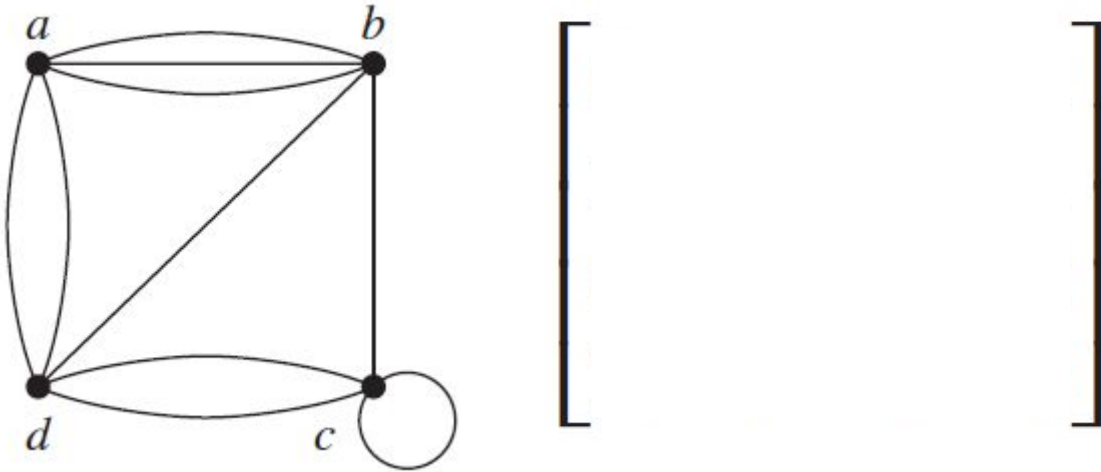
Adjacency Matrices Examples



$$\begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$



Adjacency Matrices Examples



Adjacency Matrix vs Adjacency List

	a	b	c	d
a	0	1	1	1
b	1	0	1	0
c	1	1	0	0
d	1	0	0	0

TABLE 1 An Adjacency List for a Simple Graph.

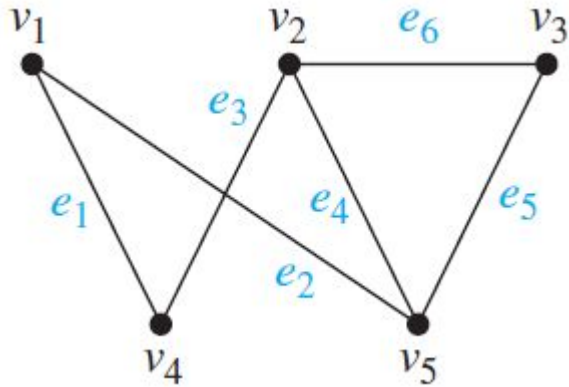
<i>Vertex</i>	<i>Adjacent Vertices</i>
<i>a</i>	<i>b, c, e</i>
<i>b</i>	<i>a</i>
<i>c</i>	<i>a, d, e</i>
<i>d</i>	<i>c, e</i>
<i>e</i>	<i>a, c, d</i>

Incidence Matrices

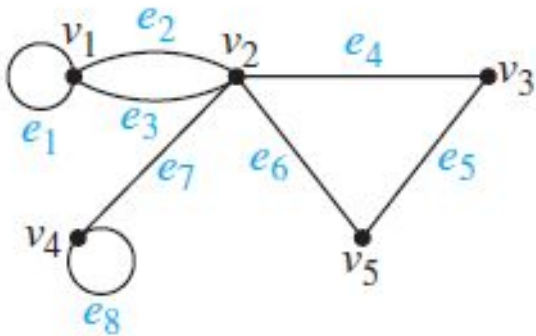
Definition 3: Let $G = (V, E)$ be an undirected graph. Suppose that $v_1, v_2, \dots, v_n$ are the vertices and $e_1, e_2, \dots, e_m$ are the edges of G . Then the incidence matrix with respect to this ordering of V and E is the $n \times m$ matrix $\mathbf{M} = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1 & \text{when edge } e_j \text{ is incident with } v_i, \\ 0 & \text{otherwise.} \end{cases}$$

Incidence Matrices Examples



$$\begin{matrix} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{bmatrix} \end{matrix}.$$



$$\begin{matrix} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} & 1 & 1 & & & & & \\ & 1 & 1 & & & & & \\ & 1 & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{bmatrix} \end{matrix}.$$